

A Fast and Adaptive Coin Selection Algorithm for UTXO-Based Blockchains

Author: Mobin Poursalami

Abstract

This paper introduces a novel, high-performance Coin Selection algorithm for UTXO-based blockchain systems. It combines the speed of randomized approaches with adaptive optimization to consistently minimize transaction fees while maintaining exceptional processing speed - outperforming traditional methods such as Branch and Bound and Knapsack in real-world applications.

1. Introduction

Unspent Transaction Outputs (UTXOs) are the backbone of Bitcoin and similar blockchain systems. Efficiently selecting UTXOs to fund transactions directly impacts transaction cost (fee) and confirmation speed. Traditional methods either sacrifice performance for precision or lack optimal fee minimization. Our proposed algorithm bridges this gap with an intelligent randomized approach.

2. Problem Definition

Given a list of UTXOs with specific amounts and a transaction target amount, the goal is to select a minimal subset of UTXOs such that:

- Their sum is greater than or equal to the target.
- The number of UTXOs is minimal (to reduce size and fee).
- The change (leftover amount) is minimal to reduce wallet fragmentation.

3. Related Work

Several coin selection strategies have been explored:

- **Greedy Algorithms:** Fast but often suboptimal, leading to higher change outputs.
- **Branch and Bound (BnB):** Accurate but computationally expensive at scale.
- **Knapsack Optimization:** Offers a theoretically sound approach, but impractical for high-throughput wallets.

Our algorithm strikes a balance between these approaches—offering speed, scalability, and effective fee minimization through adaptive and randomized techniques.

4. Proposed Method

The algorithm operates in four main stages:

- **a) Direct Match Check:** If any UTXO exactly matches the target, that UTXO is immediately selected.
- **b) Total Equality Check:** If the total sum of all available UTXOs is exactly equal to the target, all UTXOs are selected without further processing.
- **c) Insufficient Funds Check:** If the available funds are less than the target, the algorithm returns an empty set.
- **d) Randomized Best-Effort Selection:** Repeat the selection process up to N times (e.g., 1000 iterations). In each iteration, random UTXOs are selected until the sum meets or exceeds the target.

5. Algorithm Pseudocode

```
function selectUTXOs(utxos, target):
    if any u.amount == target:
        return [u]
    if sum(utxos) == target:
        return utxos
    if sum(utxos) < target:
        return []

    bestSet = None
    for i from 1 to N:
        shuffle(utxos)
        selected = []
        sum = 0
        while sum < target:
            pick = random(utxos)
            selected.append(pick)
```

```
        utxos.remove(pick)
        sum += pick.amount

    if bestSet is None or isBetter(selected, bestSet):
        bestSet = selected

return clean(bestSet, target)
```

6. Evaluation

Compared to standard methods, our algorithm:

- Is significantly faster than BnB and DP-based solutions.
- Performs competitively in minimizing change and input count.
- Scales linearly with input size in practice.

7. Use Case Scenarios

This algorithm is ideal for:

- High-frequency wallets
- Exchanges processing thousands of transactions per day
- Mobile wallets prioritizing speed and low fees

8. Conclusion

We introduced a fast and adaptive coin selection algorithm that harnesses the strengths of randomized sampling combined with an adaptive cleanup phase. This hybrid approach offers a practical solution for reducing transaction fees and processing time compared to traditional deterministic methods.

9. Future Work

Future enhancements may include:

- Enhanced Hybridization: Incorporating localized deterministic optimizations to further refine the candidate output.

- **Dynamic Iteration Adaptation:** Adjusting the number of iterations in real-time based on UTXO pool analysis for sustained optimal performance.
- **Extended Benchmarking:** Carrying out comprehensive performance evaluations across multiple blockchain platforms and wallet implementations.

10. Author Biography

Mobin Poursalami is an experienced backend developer with over 5 years of expertise in PHP, Laravel, REST API development, blockchain solutions, and coin selection algorithms. With proven experience in implementing robust solutions like the algorithm described in this paper—including its successful integration into the Xapa digital currency wallet—Mobin brings both technical insight and practical industry experience to advanced blockchain applications.